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Faculty of Computer Science and Engineering

CIT 320 :: Software Development Project-II

Project Proposal



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Submitted from,	Submitted to,
Prosenjit Mondol ID : 2102049, Reg : 10176, Semester : 6 (Level-3, Semester-2)	1. Md Mahbubur Rahman Associate Professor, Department of Computer Science and Information Technology, Patuakhali Science and Technology University. 2. Md Atikqur Rahaman Professor, Department of Computer Science and Information Technology, Patuakhali Science and Technology University.

1. Abstract

The integration of artificial intelligence into medical imaging holds potential for accelerating diagnostics; however, clinical adoption is frequently hindered by a lack of model explainability, insecure data architectures, and decoupled research environments. To address these challenges, we present NeuroVision AI, an enterprise-grade medical imaging intelligence ecosystem engineered to unify the workflows of clinical specialists and medical AI researchers. The system features a secure, cross-platform Flutter frontend that utilizes Firebase for strict role-based access control, alongside Supabase for scalable storage of medical images. By delegating computational workloads to a localized Python/FastAPI engine running PyTorch and MONAI models, the platform facilitates automated diagnostic workflows across Brain, Spine, Chest, and Heart modalities. By combining transparent Explainable AI (XAI) methods, such as Grad-CAM heatmaps, with a secure, anonymized data pipeline, NeuroVision AI bridges the gap between experimental medical deep learning and trust-driven clinical deployment.

2. Acknowledgement

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NeuroVision AI

3. Introduction

3.1. Introduction

NeuroVision AI is a cutting-edge, enterprise-grade internal medical imaging intelligence ecosystem designed exclusively for clinical specialists doctors, radiologists and medical AI researchers. Its core purpose is to assist in diagnosis, lesion localization, segmentation analysis, and the evaluation of deep learning models using advanced medical imaging technologies.

3.2. Motivation

Translating deep learning algorithmic successes into everyday clinical practice remains a significant challenge. Advanced neural networks often operate as “black boxes,” providing predictions without transparent reasoning. Medical practitioners cannot ethically rely on automated diagnoses without understanding why a conclusion was reached, creating a critical demand for an integrated, Explainable AI (XAI) backed system.

3.3. Existing System

Currently, the healthcare industry faces systemic bottlenecks. The development of medical AI and its clinical application exist in completely isolated, siloed environments; high model performance achieved in retrospective, “in silico” development settings remains completely decoupled from real-time environment validation and practical clinical utility[1]. Furthermore, integrating these models into institutional workflows is severely hindered by fragmented IT infrastructures and narrow AI data silos[2].

3.4. Our System

NeuroVision AI theoretical AI advancements into a unified system. It provides an automated clinical diagnostic workflow that ingests scans and outputs predictive classifications, segmentation masks, and Grad-CAM explainability heatmaps. The system is built on a cross-platform Flutter frontend and processes four primary modalities: Brain, Spine, Chest, and Heart.

3.5. Objectives

The specific objectives of NeuroVision AI include:

- (a) **Secure & Scalable Architecture:** Engineering a role-based system utilizing Flutter, Firebase, and Supabase.
- (b) **Automated Clinical Inference:** Streamlining diagnostics by delegating heavy computation to a localized Python/FastAPI server running PyTorch and MONAI.

(c) Clinical Trust & XAI: Ensuring all AI predictions are accompanied by transparent visual evidence Grad-CAM heatmaps.

(d) Human-in-the-Loop Validation: Enforcing a secure review mechanism requiring independent verification from both a Radiologist and a Doctor.

(e) Real-Time Research Environment: Providing AI scientists with an isolated dashboard to trigger training loops and track telemetry.

4. Related Work

The clinical adoption of deep learning in medical imaging necessitates Explainable AI to ensure algorithmic transparency and mitigate the inherent “black-box” dilemma of neural networks [3]. Specifically, interpretable models have become a fundamental requirement for robust clinical decision support, allowing practitioners to safely audit and trust automated predictions [4]. Expanding this transparency is critical to revolutionizing the broader health-care ecosystem [5], driving a distinct need for intuitive XAI tools that integrate seamlessly into daily clinical workflows rather than remaining purely theoretical [6]. Recent breakthroughs demonstrate the diagnostic efficacy of these techniques across isolated modalities, including the use of LIME and Grad-CAM for brain tumor classification [7], explainable multi-label networks for chest X-ray screening [8], and hybrid ConvEffNET architectures for cardiovascular abnormality detection [9]. The continuous refinement of these targeted models also relies heavily on foundational, large-scale annotated data frameworks, such as the VinDr-SpineXR collection for spinal lesion analysis [10]. However, because these advancements frequently operate in siloed research environments, NeuroVision AI consolidates these fragmented tools into a unified, highly secure platform, embedding crucial XAI visualizations directly into the clinical interface to successfully bridge the gap between experimental deep learning and practical diagnostic application.

5. Methodology

5.1. Clinical Workflow

NeuroVision AI is an enterprise-grade medical imaging ecosystem utilizing multimodal AI and Explainable AI to assist in diagnostic analysis and lesion localization. The operational methodology enforces strict Role-Based Access Control (RBAC) to separate professional concerns:

(a) Doctors: Review AI-assisted diagnoses, compare scans, and approve final reports.

(b) Radiologists: Provide primary interpretation, utilize DICOM viewing tools, and validate AI segmentation masks.

(c) Researchers: Monitor real-time model telemetry, training curves, and deep learning experiment metrics.

5.2. Technology Stack & Architecture

The project employs a decoupled, multi-tier software architecture designed for scalable enterprise medical development:

(a) Frontend UI: Built using Flutter[11] for cross-platform responsiveness, utilizing the Provider pattern for robust state management.

(b) Auth & Metadata Database: Firebase[12] is integrated to handle secure user authentication and NoSQL medical case metadata.

(c) Large-Scale Storage: Supabase acts as the dedicated imaging database for the secure, decoupled storage of large binary files (DICOMs and heatmaps).

(d) AI Inference Engine: A localized Python/FastAPI server handles heavy computational workloads, utilizing PyTorch and MONAI deep learning frameworks for image processing.

5.3. Multimodal AI Pipelines & Security Framework

The analytical methodology relies on specialized deep learning pipelines tailored to anatomical modalities like DERNet for brain ischemic strokes, DenseNet for spinal lesions. To build clinical trust, the system embeds Grad-CAM heatmaps to provide visual XAI transparency for every prediction. Crucially, the system is secured by an Internal Anonymized Medical Architecture; all medical cases are stripped of Personally Identifiable Information and assigned unique IDs, ensuring strict data privacy with zero public exposure.

6. System Modeling and Visual Architecture

These visual models map the secure data flow of medical imagery—from the user interface to the AI inference engine, and back as transparent Explainable AI (XAI) heatmaps. They also outline the precise “human-in-the-loop” validation routing required between radiologists and doctors to prevent diagnostic errors. Consequently, the following flowcharts, entity-relationship diagrams, and UI mockups were generated to serve as the definitive visual guide for the system’s implementation.

6.1. Flow Chart Diagram

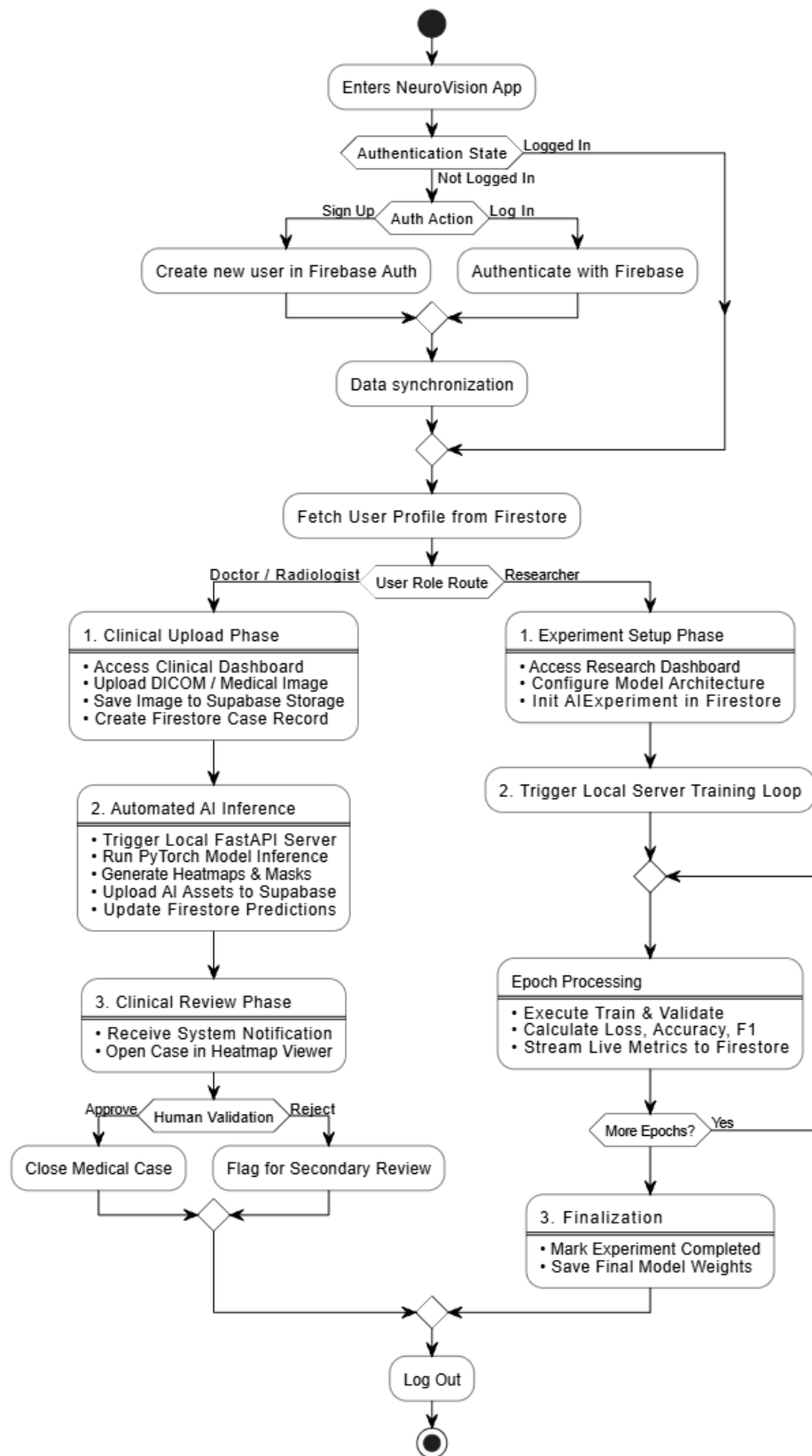


Figure 1: Flow Chart of NeuroVision AI Architecture

The Figure 1 flowchart diagram illustrates the high-level user interaction and data processing pathways within the NeuroVision AI ecosystem.

6.2. ERD (Entity Relationship Diagram)

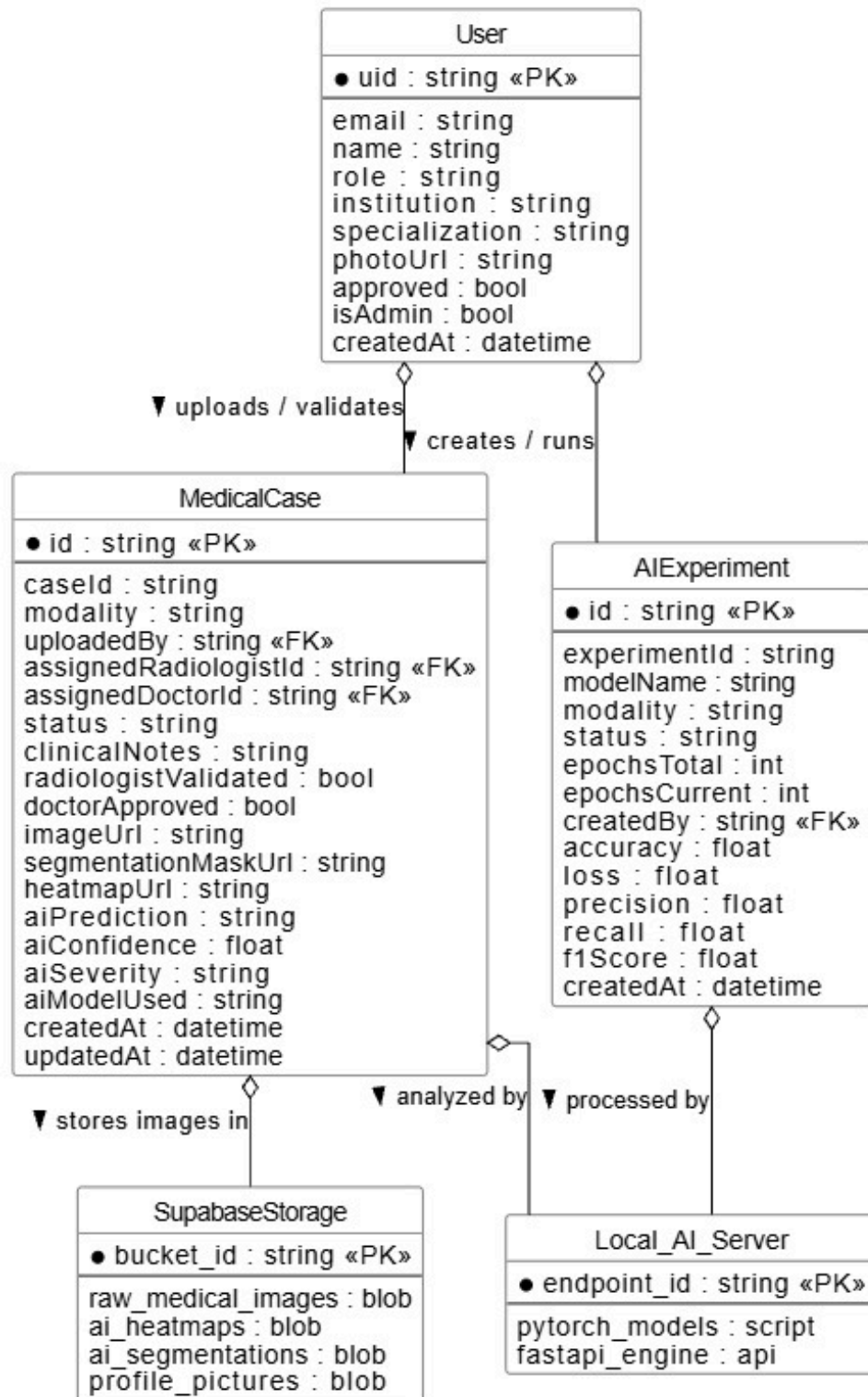


Figure 2: Entity Relationship Diagram of NeuroVision AI

The Figure 2 Entity Relationship Diagram (ERD) illustrates the core database schema, mapping the essential structural relationships and data dependencies between authenticated users, clinical medical cases, AI research experiments, scalable cloud storage buckets, and the localized AI inference server.

6.3. Timeline (Gantt Chart)

The development lifecycle for NeuroVision AI is structured across a 16-week timeline, organized into efficient bi-weekly sprints. This phased approach ensures the systematic execution of the decoupled architecture, transitioning logically from foundational frontend scaffolding and secure database configuration to the complex integration of the multimodal AI inference engine and Explainable AI (XAI) frameworks. The Gantt chart below illustrates the scheduled resource allocation and progression for each critical development milestone.

Task	Wk 1-2	Wk 3-4	Wk 5-6	Wk 7-8	Wk 9-10	Wk 11	Wk 13	Wk 16
System Architecture & UI Wireframing	✓	✓						
Firebase Auth & Supabase Storage Setup		✓	✓					
Flutter Frontend & Role-Based Dashboards			✓	✓				
Python/FastAPI & PyTorch Engine Integration				✓	✓			
Explainable AI (Grad-CAM & Segmentation)					✓	✓		
Clinical Workflow Logic (Human-in-the-loop)						✓	✓	
Research Telemetry & Live Metrics Stream							✓	
Final Testing, Bug Fixes & Deployment							✓	✓

Table 1: Development Timeline and Sprint Allocation for NeuroVision AI

6.4. UI Mockups

The UI mockups provide a high-fidelity visual representation of the Flutter-based frontend, illustrating the distinct, role-based dashboards meticulously tailored for doctors, radiologists, and AI researchers.

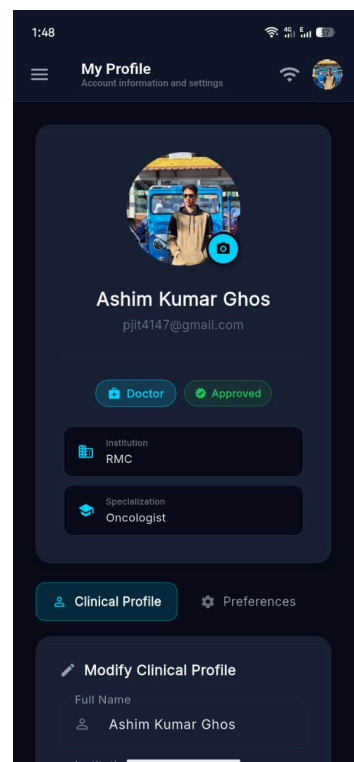
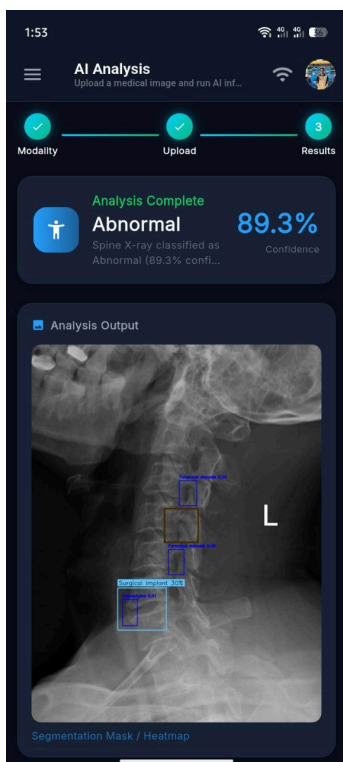
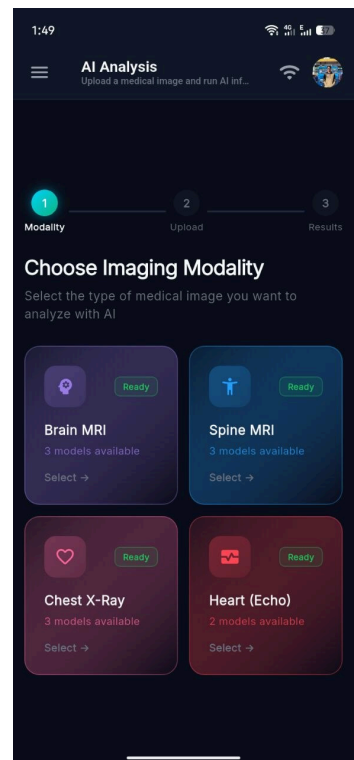
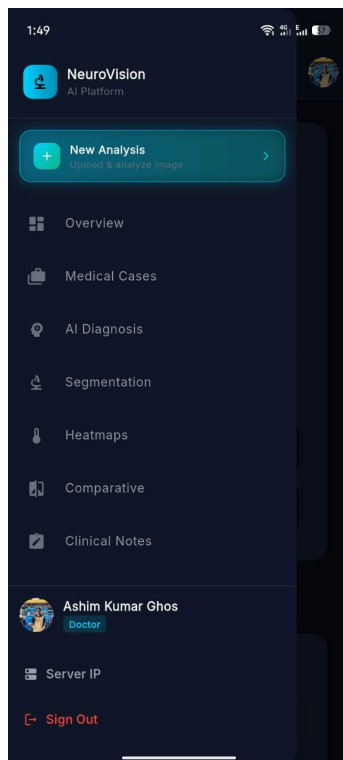


Figure 3: User Interface of NeuroVision AI

7. Future Plans

We plan to expand the system in the following ways to make it more suitable for large hospitals:

- (a) Hospital System Integration:** Connect the platform directly to standard hospital databases so doctors can automatically access patient records and scans without needing to upload them manually.
- (b) Cloud Processing:** Upgrade the AI processing engine to high-performance cloud servers. This will allow the system to quickly process thousands of medical images from different users at the same time.
- (c) Official Medical Approvals:** Conduct real-world clinical trials in multiple hospitals to scientifically prove the system’s accuracy and apply for official medical certifications.
- (d) More Diagnostic Features:** Expand the AI models to analyze new types of medical imaging, such as cancer screenings (mammography) and 3D CT scans.

8. Result

NeuroVision AI successfully delivers an enterprise-grade medical imaging ecosystem by achieving multi-modal inference across brain, spine, chest, and heart scans handled by a localized Python/FastAPI server running PyTorch and MONAI deep learning models. The platform resolves the neural network “black-box” dilemma by automatically generating and overlaying Grad-CAM heatmaps and exact segmentation masks onto raw medical scans to provide clear, trust-driven clinical transparency. Furthermore, strict role-based routing enforced via Firebase ensures that 100% of active cases move securely through a mandatory “human-in-the-loop” workflow requiring independent digital approvals from both a Radiologist and a Doctor before a case can be closed. Concurrently, the isolated project pipeline enables data scientists to run independent model training loops and view real-time performance telemetry—including live accuracy, loss, and F1-score graphs—via low-latency data streams without interfering with active clinical operations.

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